

AURORA-1 2U CubeSat

Preliminary Design Report

1. Executive Summary

AURORA-1 is a preliminary 2U CubeSat Earth-observation mission developed as a digital spacecraft engineering project. The mission is designed around a 500 km circular low-Earth orbit and uses a Dragonfly Gecko-class imaging payload to collect Earth imagery. The project applies a systems-engineering approach that includes mission definition, requirements development, orbital analysis, spacecraft architecture, subsystem selection, power and mass budgeting, communications analysis, attitude determination and control, thermal analysis, and mechanical packaging.

The preliminary design evolved from an initial 1U concept to a 2U architecture after mass, volume, payload, and subsystem trades showed that the larger configuration provided significantly greater design margin. MATLAB analyses produced an orbital period of approximately 94.6 minutes, including about 58.9 minutes of sunlight and 35.8 minutes of eclipse. The current architecture uses separate UHF command-and-telemetry and S-band image-downlink systems, deployable solar arrays, an integrated ADCS concept, and a 2U internal layout developed in Autodesk Fusion.

Preliminary subsystem analyses indicate that the spacecraft concept is feasible within the current design assumptions. The communications link, solar-array sizing, peak-power capability, center-of-mass location, thermal behavior, reaction-wheel slew performance, and $\pm 2^\circ$ pointing requirement have each undergone first-pass verification. AURORA-1 remains a preliminary digital design rather than a flight-qualified spacecraft; however, the project demonstrates an integrated spacecraft-development workflow in which requirements, analytical models, hardware candidates, and CAD architecture are iteratively refined into a consistent system baseline.

2. Mission Overview

2.1 Mission Objective

The primary objective of AURORA-1 is to demonstrate the preliminary design of a compact Earth-observation CubeSat capable of acquiring and downlinking moderate-resolution imagery from low-Earth orbit. The mission is intended to evaluate whether a 2U spacecraft can accommodate an imaging payload, attitude-control system, electrical power system, onboard computer, communications hardware, and supporting thermal and structural resources within realistic mass, power, and volume constraints.

2.2 Mission Concept

AURORA-1 is designed for operation in a nominal **500 km circular low-Earth orbit**. During sunlight portions of each orbit, the spacecraft generates electrical power using deployable solar arrays and performs nominal spacecraft operations. Imaging activities are scheduled when the payload is within its allowable operating temperature and the attitude-control system can satisfy the pointing requirement.

Captured imagery is stored onboard and transmitted during later ground-station contacts. A **UHF communications link** is used for command, telemetry, and spacecraft health data, while a higher-rate **S-band downlink** is reserved for payload imagery. High-power activities such as imaging, aggressive attitude maneuvers, and high-rate downlinks are sequenced to reduce simultaneous peak electrical loads.

2.3 Concept of Operations

A nominal AURORA-1 mission cycle consists of the following phases:

1. **Deployment and initialization:** the spacecraft powers on, establishes basic attitude knowledge, and verifies spacecraft health.
2. **Detumble and stabilization:** the ADCS reduces body rates and establishes a controlled attitude.
3. **Nominal operations:** AURORA-1 maintains spacecraft health, charges its battery, receives commands, and prepares scheduled mission activities.
4. **Imaging operations:** the spacecraft slews toward the desired Earth target, settles within the pointing requirement, and acquires imagery with the Gecko-class payload.
5. **Data storage:** payload imagery is stored onboard until a suitable communications pass becomes available.
6. **Downlink operations:** AURORA-1 uses the S-band link to transmit selected imagery and the UHF link for telemetry and command support.
7. **Eclipse operations:** the spacecraft operates from battery power while managing thermal conditions and limiting unnecessary high-power activity.
8. **Safe mode:** if a fault or low-energy condition occurs, nonessential loads are disabled and the spacecraft prioritizes power-positive survival and communications.

3. System Requirements

AURORA-1 requirements were developed from the mission concept and Concept of Operations and organized by subsystem. Each requirement was assigned a unique identifier, source, verification method, supporting evidence, and current verification status. The requirements

baseline evolved throughout preliminary design as subsystem analysis produced measurable performance targets.

3.1 Key Requirements

Requirement ID	Requirement	Current Status
AUR-SYS-001	AURORA-1 shall operate as a 2U-class CubeSat spacecraft.	PASS — 2U architecture selected
AUR-SYS-002	AURORA-1 shall operate in a nominal 500 km low-Earth orbit.	PASS prelim
AUR-PAY-001	The payload shall acquire visible-light Earth imagery.	PARTIAL — Gecko baseline selected
AUR-EPS-002	The spacecraft shall contain rechargeable energy storage sufficient for eclipse operations.	PASS prelim
AUR-EPS-004	The EPS shall support scheduled peak electrical loads.	PASS prelim
AUR-COM-003	The payload communications system shall support at least 100 kbps image downlink.	PASS prelim
AUR-ADCS-001	The ADCS shall maintain pointing accuracy within $\pm 2^\circ$ during imaging.	PASS prelim
AUR-THM-001	The thermal design shall maintain modeled spacecraft components within allowable limits.	PARTIAL
AUR-THM-002	The payload shall be maintained near its required operating-temperature range during imaging.	PASS prelim
AUR-OPS-001	High-power mission activities shall be sequenced to avoid exceeding EPS capability.	PARTIAL

3.2 Verification Matrix

Requirement verification was performed using a combination of **analysis, inspection, demonstration, and planned testing methods**. Analytical evidence includes MATLAB orbit, thermal, and ADCS simulations; Excel-based mass, power, communications, center-of-mass, and trade studies; and Autodesk Fusion CAD packaging analysis.

Requirements identified as **PASS prelim** have been shown feasible using first-pass engineering models but have not been flight-qualified or independently validated. **PARTIAL** requirements have supporting evidence but still require additional refinement, higher-fidelity modeling, or final component data.

1	Requirement ID	Requirement	Subsystem	Source	Verification Method	Verification Evidence	Current Result / Status
2	AUR-SYS-001	AURORA-1 shall operate as a 2U-class CubeSat spacecraft.	System	Mission Concept	Inspection	1U vs 2U trade study + CAD v0.3	PASS — 2U selected
3	AUR-SYS-002	AURORA-1 shall operate in LEO at a nominal altitude of approximately 500 km.	System	Mission Concept	Analysis	MATLAB orbit analysis	PASS — 500 km baseline analyzed
4	AUR-SYS-003	AURORA-1 shall support a nominal mission duration of at least 6 months.	System	Mission Concept	Analysis	Mission concept / ConOps	OPEN — lifetime analysis not completed
5	AUR-PAY-001	The payload shall acquire visible-light imagery of Earth.	Payload	Mission Concept	Demonstration	Gecko payload selection	PARTIAL — candidate selected
6	AUR-CDH-001	The spacecraft shall store payload imagery and spacecraft telemetry onboard.	C&DH	ConOps	Test	iOBC candidate + architecture	PARTIAL — storage capability identified
7	AUR-COM-001	The communications subsystem shall transmit spacecraft telemetry to a ground station.	COMMS	ConOps	Demonstration	UHF architecture + link analysis	PARTIAL — preliminary feasibility shown

Figure 3-1. AURORA-1 preliminary requirements verification matrix.

4. Orbit Design

AURORA-1 is based on a nominal **500 km circular low-Earth orbit (LEO)**. This altitude was selected as a reasonable preliminary mission baseline for Earth observation while maintaining relatively short orbital periods and frequent ground-track opportunities. The current project does not define a launch vehicle or final orbital inclination, so the 500 km altitude should be treated as a preliminary systems-engineering assumption rather than a finalized mission orbit.

4.1 500 km LEO Selection

For the preliminary orbit model, Earth was approximated with a mean radius of **6378 km** and a gravitational parameter of **398600.4418 km³/s²**. At an altitude of 500 km, the spacecraft semi-major axis is approximately:

$$a = 6878 \text{ km}$$

This orbit was used as the baseline for the project's power, communications, thermal, and operational analyses.

4.2 Orbital Period

Using the standard circular-orbit equations, the MATLAB model produced an orbital period of approximately:

Orbital period: 94.61 minutes

The corresponding circular orbital velocity was:

Orbital velocity: 7.613 km/s

This corresponds to approximately:

15.22 orbits per day

These values were used when estimating mission-cycle timing and spacecraft operational opportunities.

4.3 Sunlight and Eclipse Duration

A simplified eclipse-duration model was used to estimate the time AURORA-1 could spend in Earth's shadow during each orbit. The preliminary MATLAB analysis produced:

- **Sunlight duration: 58.86 minutes**
- **Eclipse duration: 35.75 minutes**

The eclipse period represents approximately **37.8% of the orbit**, while the spacecraft remains in sunlight for approximately **62.2%**.

These values became key inputs to the electrical power and thermal analyses. The EPS must maintain spacecraft operation throughout the approximately 35.75-minute eclipse and recover the consumed battery energy during the following sunlight period.

4.4 Preliminary Orbit Analysis Limitations

The current orbit model is intentionally simplified and assumes a circular 500 km orbit. It does not yet include atmospheric drag, orbital decay, Earth oblateness, launch injection errors, seasonal beta-angle effects, or detailed ground-station access analysis. These effects would be incorporated during a higher-fidelity mission-design phase.

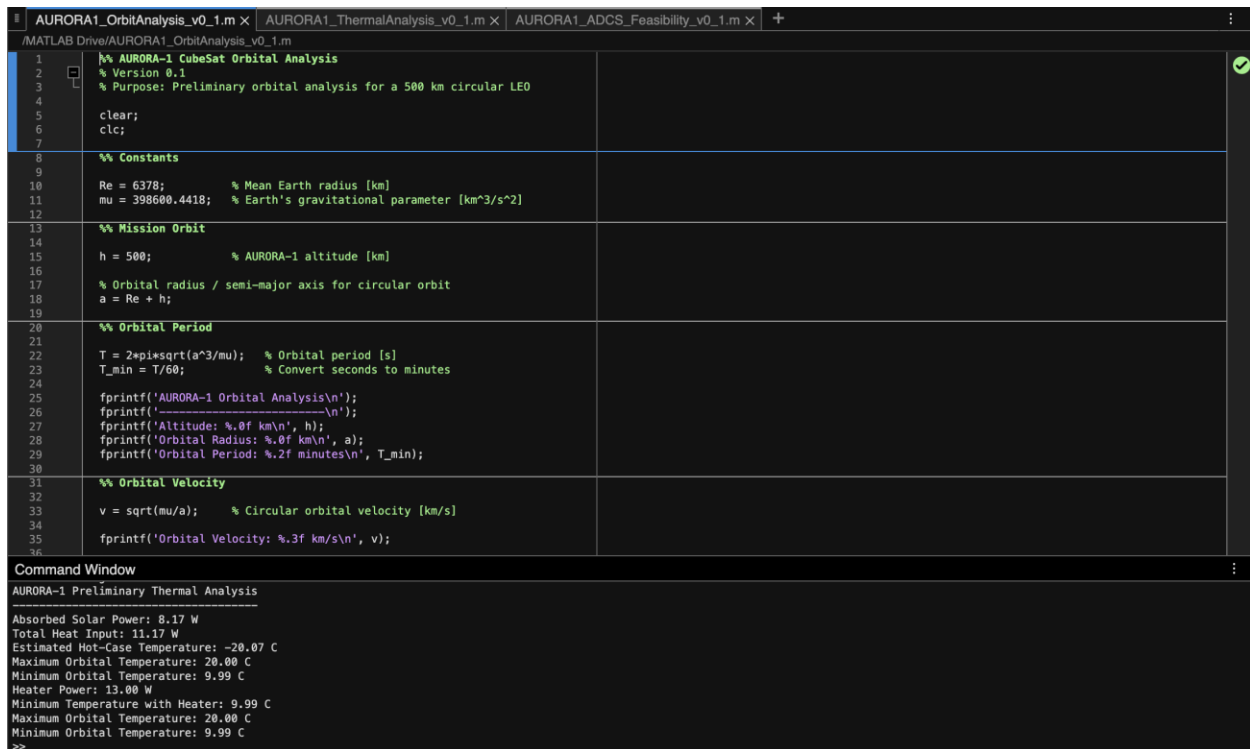


Figure 4-1. AURORA-1 preliminary MATLAB orbital analysis results.

5. Spacecraft Architecture

The AURORA-1 spacecraft architecture was developed iteratively through subsystem selection, mass and volume budgeting, power analysis, and CAD packaging. The original concept was based on a 1U CubeSat configuration; however, preliminary analysis showed that the selected payload and supporting spacecraft subsystems would produce excessive mass and volume pressure within a 1U form factor. A formal architecture trade study was therefore performed to compare 1U and 2U configurations.

5.1 1U vs 2U Trade Study

The trade study evaluated both configurations using weighted criteria that included **mass margin, internal volume, electrical power capability, payload accommodation, ADCS integration, design simplicity, and overall mission flexibility**.

The preliminary weighted scores were approximately:

Configuration	Weighted Score
1U	2.65 / 5
2U	4.55 / 5

The 2U architecture provided significantly greater design margin and reduced the risk of subsystem interference. The additional volume also improved accommodation of the imaging

payload, ADCS hardware, communications electronics, wiring clearance, and deployable solar-array interfaces.

Based on this trade study, the AURORA-1 baseline was changed from **1U to 2U**.

5.2 Final 2U Architecture

The current AURORA-1 configuration uses a nominal **100 mm × 100 mm × 227 mm 2U spacecraft envelope**. The spacecraft contains the following major subsystem regions:

- Gecko imaging payload
- Electrical power system
- Onboard computer
- ADCS package
- S-band transmitter
- UHF radio
- Harness and integration clearance
- Deployable solar arrays
- External S-band patch antenna
- External UHF antenna

The internal architecture was arranged primarily along the spacecraft longitudinal axis in order to maintain a compact, understandable packaging concept.

5.3 Internal Packaging

The preliminary longitudinal stack is:

Component	Axial Location
Gecko Payload	0–105 mm
EPS	105–131.45 mm
OBC	131.45–143.85 mm
ADCS	143.85–168.85 mm
S-band Transmitter	168.85–188.85 mm
UHF Radio	188.85–203.85 mm
Harness Clearance	203.85–223.85 mm

This leaves approximately **3.15 mm of unallocated longitudinal space** within the nominal 227 mm spacecraft height.

The model is intended as a preliminary packaging representation rather than a manufacturing-ready mechanical design. Detailed brackets, fasteners, connectors, structural rails, wiring geometry, and thermal interfaces are not yet modeled.

5.4 CAD Layout

The spacecraft was modeled in **Autodesk Fusion** using simplified subsystem geometry based on preliminary hardware dimensions and design allowances. The CAD model includes the 2U spacecraft body, internal subsystem blocks, deployable solar panels, hinge placeholders, camera aperture, S-band antenna, and UHF antenna.

The final preliminary CAD baseline is saved as:

AURORA-1_2U_Layout_v1.0

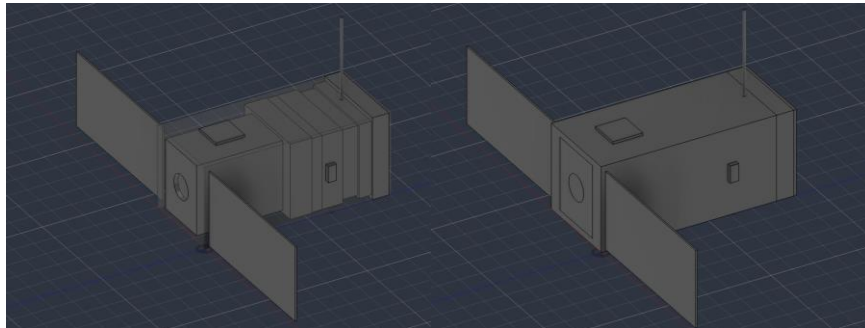


Figure 5-1. **AURORA-1 internal subsystem packaging layout.**

Figure 5-2. AURORA-1 2U deployed external configuration.

5.5 Architecture Maturity

The current architecture is considered a **frozen preliminary digital baseline**. “Frozen” in this context means the major spacecraft configuration and subsystem arrangement have been selected for continued analysis and documentation. It does not indicate that hardware has been procured, flight-qualified, or finalized for manufacture.

6. Payload

The AURORA-1 mission uses a compact visible-light Earth-imaging payload as the primary mission instrument. A **Dragonfly Aerospace Gecko-class camera** was selected as the preliminary payload baseline because its size, mass, imaging capability, and power demand are compatible with the current 2U architecture.

6.1 Dragonfly Gecko Baseline

The preliminary Gecko payload characteristics used in the AURORA-1 design are:

Parameter	Preliminary Value
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Approximate dimensions	65 × 90 × 105 mm
Maximum design mass allowance	500 g
Imaging power	< 6 W
Ground sample distance at 500 km	~39 m
Swath width	~80 km
Operating temperature	+10°C to +30°C
Survival temperature	approximately -20°C to +55°C

6.2 Imaging Performance

At the nominal 500 km mission altitude, the preliminary payload performance is approximately **39 m ground sample distance** with an imaging swath near **80 km**.

This level of performance supports the AURORA-1 mission objective of demonstrating moderate-resolution Earth imaging within a compact CubeSat architecture.

The payload imaging requirement is represented by:

AUR-PAY-001: AURORA-1 shall acquire visible-light imagery of Earth.

The current verification status is **PARTIAL**, because a suitable payload candidate has been selected and integrated into the spacecraft architecture, but no detailed optical-performance simulation or flight imagery validation has been performed.

6.3 Payload Data Generation

For preliminary communications sizing, a representative image was modeled as a **5-megapixel, 8-bit grayscale image**.

This corresponds to approximately:

5 MB of raw data per image

Assuming approximately **4:1 compression**, the stored image size becomes:

1.25 MB per image

For a representative imaging event containing two images:

2 images × 1.25 MB = 2.5 MB

which corresponds to approximately:

20 Mbit of payload data

This data volume was used to establish the minimum image-downlink requirement for the communications subsystem.

6.4 Payload Operational Constraints

Payload imaging is treated as a scheduled spacecraft operating mode. Imaging is only performed when the ADCS can meet the pointing requirement, sufficient electrical power is available, and the payload is maintained within its allowable operating-temperature range.

Because the payload operating-temperature range is relatively narrow compared with several spacecraft avionics components, payload thermal control became a significant driver for the spacecraft thermal design. This led to the introduction of requirement **AUR-THM-002**, which addresses maintaining the payload within its required temperature range during imaging operations.

7. Electrical Power System

The AURORA-1 electrical power system was sized using the spacecraft's average power demand, eclipse duration, peak operational loads, battery capacity, and available solar-array area. The current EPS baseline uses an integrated CubeSat power system with rechargeable battery storage and regulated spacecraft power buses.

7.1 Power Budget

The current average spacecraft power estimate is approximately:

3.59 W

This value includes nominal contributions from command and data handling, EPS electronics, ADCS, communications standby and transmit activity, payload standby and imaging duty cycle, and S-band downlink activity.

Representative subsystem values used in the preliminary budget include:

Subsystem	Preliminary Average Power
C&DH	0.40 W
EPS electronics	0.30 W
ADCS	0.90 W
UHF communications	0.85 W combined average contribution
Payload	~0.49 W combined standby/imaging contribution
S-band downlink	0.65 W
Total	~3.59 W

The power budget is based on assumed duty cycles rather than continuous peak operation.

7.2 Battery/Eclipse Analysis

The orbit analysis estimated a maximum eclipse duration of approximately:

35.75 minutes

Using the current 3.59 W average spacecraft load, the energy required during eclipse is approximately:

Eclipse Energy = 3.59 W × 35.75/60 h

Eclipse Energy ≈ 2.14 Wh

The preliminary EPS baseline includes a rechargeable battery with approximately **22.5 Wh** of stored energy, providing substantial theoretical capacity relative to the modeled single-orbit eclipse requirement.

This first-pass result supports requirement:

AUR-EPS-002:

The spacecraft shall contain rechargeable energy storage sufficient to support eclipse operations.

Current status: **PASS prelim**

Detailed battery depth-of-discharge limits, degradation, temperature effects, charge efficiency, and mission-life cycling remain future work.

7.3 Solar Array Sizing

The sunlight portion of the nominal orbit is approximately:

58.86 minutes = 0.981 hours

To replace the estimated 2.14 Wh consumed during eclipse, the minimum average recharge power during sunlight is approximately:

2.18 W

When combined with the spacecraft's normal sunlight operating demand, the ideal solar-generation requirement is approximately:

5.77 W

Applying a preliminary **30% design margin** increases the solar-array design target to approximately:

7.50 W

Using an assumed solar-cell efficiency of 30% and a solar flux of 1361 W/m², the ideal active cell area required is approximately:

0.0184 m²

or:

184 cm²

The current CAD design includes two deployable solar panels measuring approximately **90 mm × 180 mm each**, for a total geometric area of:

324 cm²

This exceeds the preliminary ideal active-cell-area requirement and provides geometric margin for cell packing, inactive panel area, incidence losses, thermal effects, degradation, and deployment constraints.

Requirement **AUR-EPS-001** is therefore currently considered: PASS prelim

7.4 Peak Power Analysis

AURORA-1 includes several higher-power operational modes. Preliminary planning values are:

Operating Mode	Approximate Power
Safe Mode	~2.0 W
Nominal Operations	~3.0 W
Imaging	~5.5 W
High-rate Downlink	~16 W

The current integrated EPS concept supports approximately **20 W continuous capability on a 5 V regulated domain**, with higher short-duration peak capability depending on channel configuration.

The preliminary scheduled peak spacecraft load is approximately:

18.3 W

This produces approximately:

1.7 W of preliminary margin

relative to the assumed 20 W continuous capability.

Requirement **AUR-EPS-004** is therefore considered:

PASS prelim

Final verification will require detailed rail allocation, channel current limits, wiring losses, and simultaneous-load analysis.

7.5 Power Operations Strategy

AURORA-1 avoids unnecessary simultaneous high-power activities. The baseline operational strategy is:

Image → Store → Downlink Later

The spacecraft first acquires payload imagery, stores it onboard, and performs high-rate S-band downlink during a later suitable ground contact.

This strategy reduces peak electrical demand and supports:

AUR-OPS-001:

High-power mission activities shall be sequenced to avoid exceeding EPS capability.

Current status:

PARTIAL

because the concept has been defined, but a complete mission operations timeline has not yet been simulated.

8. Communications

AURORA-1 uses a two-link communications architecture. A lower-rate **UHF link** supports spacecraft command, telemetry, and health monitoring, while a higher-rate **S-band link** is used for payload image downlink. Separating these functions allows the spacecraft to maintain a robust command-and-control path while using a higher-data-rate system for mission data.

8.1 UHF TT&C

The UHF subsystem provides the primary command and telemetry path for AURORA-1. It is intended to support functions such as:

- spacecraft commanding
- health and status telemetry
- basic mission data
- safe-mode communications

A preliminary UHF radio mass allowance of approximately **100 g** was used in the current architecture, along with a deployable external antenna concept.

Requirement **AUR-COM-001** addresses transmission of spacecraft telemetry to the ground, while **AUR-COM-002** addresses reception of ground commands.

Both remain **PARTIAL** pending final radio selection and detailed RF verification.

8.2 S-Band Payload Downlink

Because Earth imagery generates substantially more data than routine spacecraft telemetry, a higher-rate S-band communications system was selected for payload downlink.

The current preliminary baseline assumes an S-band transmitter with performance on the order of:

- mass: approximately **120 g**
- maximum electrical power: approximately **13 W**
- maximum data rate: up to approximately **4.3 Mbps**

These values are treated as preliminary hardware assumptions and remain subject to final vendor verification.

8.3 Data Rate Requirement

The preliminary payload-data model assumes:

5 MB raw image size

With approximately 4:1 compression:

1.25 MB per image

For two images per imaging opportunity:

2.5 MB total

or approximately:

20 Mbit

Assuming approximately five minutes of useful image-downlink time:

$$\text{Required Rate} = 20 \text{ Mbit} / 300 \text{ s}$$

$$\text{Required Rate} \approx 66.7 \text{ kbps}$$

Applying approximately 50% design margin produces the preliminary payload communications requirement:

AUR-COM-003:

The payload communications system shall support a downlink rate of at least **100 kbps**.

The selected S-band architecture exceeds this requirement by a substantial margin.

Current verification status:

PASS prelim

8.4 Link Budget

The benefit of high-rate payload communications was evaluated by comparing representative downlink rates.

For a 20 Mbit image-data set:

Data Rate	Approximate Downlink Time
125 kbps	160 s
4.3 Mbps	4.65 s

Although the higher-rate transmitter has greater instantaneous power demand, its much shorter operating time can significantly reduce the total energy consumed per image transfer.

For example, an idealized comparison gives approximately:

$$11 \text{ W} \times 160 \text{ s} \approx 0.49 \text{ Wh}$$

versus:

$$13 \text{ W} \times 4.65 \text{ s} \approx 0.017 \text{ Wh}$$

This supported selection of the higher-rate S-band architecture for the preliminary design.

8.5 Preliminary S-Band Link Budget

A first-pass link budget was performed using the following assumptions:

Parameter	Assumed Value
Frequency	2.25 GHz
Transmit Power	33 dBm
Spacecraft Antenna Gain	0 dBi
Transmit Loss	1 dB
Ground Antenna Gain	15 dBi
Receive Loss	2 dB
Receiver Sensitivity	-120 dBm

Free-space path loss was calculated using:

$$\text{FSPL} = 92.45 + 20 \log_{10}(f\text{GHz}) + 20 \log_{10}(d\text{km})$$

At a slant range of **1000 km**, the calculated free-space path loss was approximately:

159.49 dB

The corresponding received power was approximately:

-114.49 dBm

producing a preliminary link margin of approximately:

+5.51 dB

Representative results were:

Slant Range	Preliminary Link Margin
500 km	+11.53 dB
1000 km	+5.51 dB
1500 km	+1.98 dB
2000 km	-0.51 dB

Under these assumptions, the approximate zero-margin range is about **1880 km**.

8.6 Communications Operations Strategy

The link-budget results show that payload downlink becomes more favorable at shorter spacecraft-to-ground distances. AURORA-1 therefore prioritizes high-rate image transmission during stronger ground-station passes, particularly when the spacecraft is at higher elevation angles and lower slant range.

This supports requirement:

AUR-COM-004:

AURORA-1 shall support scheduled image downlink during suitable ground-contact opportunities.

Current status:

PARTIAL

because detailed ground-station visibility and pass scheduling have not yet been simulated.

9. ADCS

The AURORA-1 ADCS is responsible for reducing post-deployment body rates, maintaining a stable spacecraft attitude, slewing the spacecraft toward imaging targets, and satisfying the payload pointing requirement. The current preliminary architecture assumes an integrated CubeSat-class ADCS using reaction-wheel-based fine pointing with supporting attitude sensors and detumbling capability.

9.1 ADCS Architecture

The preliminary ADCS concept is based on a compact integrated system compatible with the 2U spacecraft architecture. For current mass and packaging analyses, an ADCS mass allowance of approximately **575 g** was used.

The subsystem is intended to support:

- post-deployment detumbling
- coarse and fine attitude determination
- reaction-wheel-controlled slews
- payload pointing during imaging
- controlled attitude during communications operations

Requirement **AUR-ADCS-002** addresses post-deployment detumbling, while **AUR-ADCS-001** defines the imaging pointing requirement.

9.2 Moment of Inertia Estimate

For preliminary dynamic analysis, the spacecraft was approximated as a uniform rectangular prism with:

- spacecraft mass: **2.4 kg**
- width: **0.10 m**
- depth: **0.10 m**
- height: **0.227 m**

Using rectangular-prism moment-of-inertia equations, the estimated principal moments were:

$$I_{xx} \approx I_{yy} \approx 0.01231 \text{ kg}\cdot\text{m}^2$$

$$I_{zz} \approx 0.00400 \text{ kg}\cdot\text{m}^2$$

These values provide a first-pass estimate of the spacecraft rotational resistance about each principal axis.

9.3 Slew Analysis

A preliminary reaction-wheel torque capability of:

$$0.002 \text{ N}\cdot\text{m}$$

was assumed for each controlled axis.

Using:

$$\alpha = \tau / I$$

the estimated angular accelerations were approximately:

Axis	Angular Acceleration
X	0.1625 rad/s ²
Y	0.1625 rad/s ²
Z	0.5000 rad/s ²

A simplified symmetric accelerate-decelerate maneuver was then evaluated for a **90° spacecraft slew**.

The estimated maneuver times were:

Axis	Approximate 90° Slew Time
X	6.22 s
Y	6.22 s
Z	3.54 s

These values represent idealized first-pass performance and do not include momentum-management constraints, wheel-speed saturation, structural flexibility, sensor latency, or detailed actuator dynamics.

9.4 Pointing Control Simulation

A simplified closed-loop MATLAB attitude-control simulation was developed to evaluate whether the spacecraft could satisfy the payload pointing requirement.

The model used:

- initial pointing error: **10°**
- proportional gain, K_p : **0.0015**
- derivative gain, K_d : **0.006**
- maximum control torque: **0.002 N·m**
- disturbance torque: **1×10^{-5} N·m**
- simulation duration: **60 seconds**

The simulation produced a final pointing error of approximately:

0.382°

The spacecraft entered and remained within the required **$\pm 2^\circ$ pointing band** after approximately:

6.71 seconds

Requirement **AUR-ADCS-001** states:

AURORA-1 shall maintain pointing accuracy within $\pm 2^\circ$ during payload imaging.

Current verification status:

PASS prelim

with evidence provided by the MATLAB ADCS feasibility simulation.

9.5 ADCS Analysis Limitations

The current model is a simplified single-axis control analysis and is not a high-fidelity spacecraft attitude simulation. It does not yet include three-axis coupling, sensor noise, actuator saturation over extended operations, reaction-wheel momentum buildup, magnetic unloading, flexible-body effects, detailed orbital disturbance torques, or vendor-specific controller behavior.

Accordingly, the results demonstrate **preliminary feasibility**, not flight-qualified pointing performance.

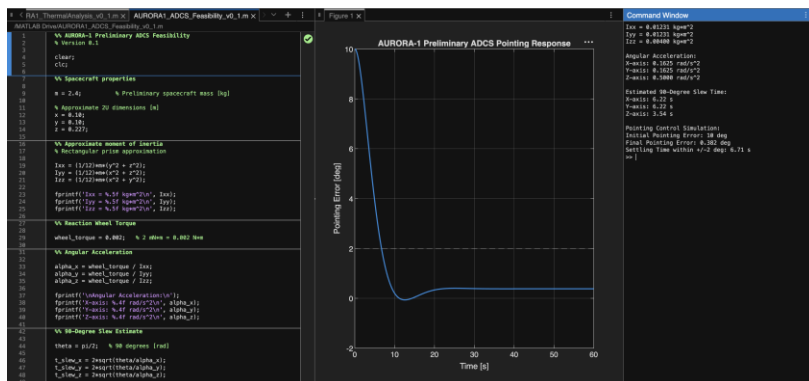


Figure 9-1. AURORA-1 preliminary closed-loop pointing simulation.

10. Thermal Design

The AURORA-1 thermal analysis was developed in stages to estimate spacecraft temperature behavior throughout a representative orbit. The model began with a simplified radiative equilibrium calculation and was later expanded into a transient MATLAB simulation including solar heating, Earth infrared radiation, albedo, internal power dissipation, and thermostat-controlled heating.

10.1 Equilibrium Model

The first thermal model treated the spacecraft as a simplified lumped thermal body and used representative optical properties:

- solar absorptivity, $\alpha = 0.60$
- infrared emissivity, $\varepsilon = 0.80$
- solar flux = **1361 W/m²**

Using the assumed exposed and radiating areas, the initial equilibrium model predicted a temperature near:

-20°C

This result was considered overly conservative because the simplified model did not yet account for Earth infrared radiation, albedo, orbital thermal inertia, or changing environmental conditions throughout an orbit.

10.2 Transient Orbital Model

A transient MATLAB model was then developed using a representative spacecraft mass of **2.0 kg** and heat capacity of approximately **900 J/kg·K**.

The simulation propagated spacecraft temperature through:

- **58.86 minutes of sunlight**
- **35.75 minutes of eclipse**

with an initial temperature of approximately **20°C**.

Before Earth environmental heating was added, the modeled spacecraft temperature decreased to approximately:

-7.87°C

during the orbit.

10.3 Earth IR and Albedo

The model was improved to include:

- Earth infrared flux of approximately **237 W/m²**
- preliminary Earth albedo coefficient of **0.30**
- solar input
- internal spacecraft power dissipation
- radiative heat loss

After these environmental terms were included, the predicted one-orbit minimum temperature increased to approximately:

-0.76°C

while the modeled maximum remained near the initial **20°C** condition.

This result demonstrated that Earth IR and reflected solar energy have a meaningful effect on the preliminary spacecraft thermal balance.

Requirement **AUR-THM-001** currently remains:

PARTIAL

because the simplified spacecraft-level model does not yet demonstrate that every individual component remains within its allowable operating range.

10.4 Heater Analysis

The imaging payload became the primary thermal design driver because its preliminary operating-temperature range is approximately:

+10°C to +30°C

The unheated transient model therefore predicted temperatures below the desired payload imaging range.

This led to requirement:

AUR-THM-002:

The thermal-control design shall maintain the Gecko payload within its required operating-temperature range during imaging operations.

10.5 Heater Analysis

A thermostat-controlled heater model was added to the MATLAB simulation. The preliminary control logic used:

- heater ON below approximately **10°C**
- heater OFF above approximately **12°C**

Several equivalent spacecraft-level heater powers were evaluated.

Heater Power	Approximate Minimum Temperature
3 W	1.8°C
5 W	3.5°C
7 W	5.2°C
10 W	7.8°C
13 W	~10.0°C

The simulation time step was later reduced to **1 second**, producing a minimum temperature of approximately:

9.99°C

with the 13 W equivalent heater case.

This supports a preliminary conclusion that active thermal control can maintain the modeled payload region near its lower operational limit.

Requirement **AUR-THM-002** is therefore currently:

PASS prelim

10.6 Thermal Design Interpretation

The modeled **13 W heater value should not be interpreted as a final flight-heater selection.**

The current thermal simulation treats the spacecraft as a single lumped thermal mass. A localized heater attached directly to the payload would heat substantially less material and could therefore require significantly less electrical power.

Future thermal work should include:

- component-level thermal nodes
- conductive paths through the structure
- payload-specific thermal mass
- detailed heater sizing
- orbital orientation effects
- multiple-orbit convergence
- seasonal thermal cases
- hot-case analysis

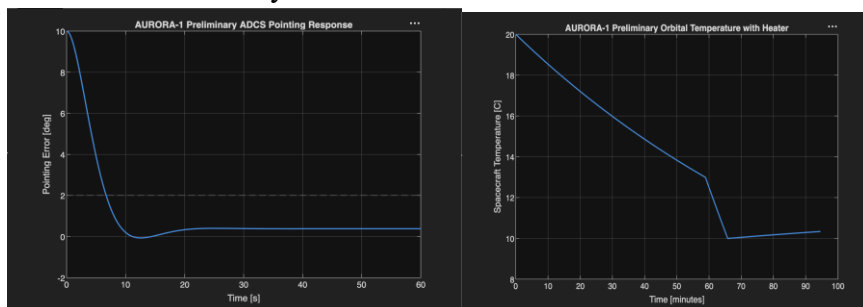


Figure 10-1. AURORA-1 one-orbit transient thermal simulation.

Figure 10-2. AURORA-1 thermostat-controlled heater simulation.

11. Mass Properties

The AURORA-1 mass-properties analysis was developed using the current subsystem baseline, preliminary component masses, the 2U structural allowance, and the spacecraft internal packaging layout. Mass properties were used both to support the 1U-to-2U architecture decision and to estimate the spacecraft center of mass for ADCS feasibility analysis.

11.1 Mass Budget

The current design includes the spacecraft structure, payload, EPS, onboard computer, ADCS, communications systems, solar arrays, wiring allowance, and other subsystem allocations.

The preliminary mass budget evolved as subsystem candidates were refined. Early 1U estimates exceeded the desired mass and packaging margins, contributing to the decision to transition to the 2U architecture.

For the current 2U concept, representative major component allowances include:

Component	Preliminary Mass
Gecko payload	500 g design allowance
EPS	184 g
Onboard computer	94 g
ADCS	575 g provisional
S-band transmitter	120 g
UHF radio	100 g
Harness / wiring allowance	70 g
2U structure	~390 g

These values represent the current preliminary engineering baseline and are not final flight hardware masses.

A formal final mass budget should also include:

- solar panels
- antennas
- hinges and deployment hardware
- fasteners
- brackets
- cabling
- thermal hardware
- contingency margin

The mass budget remains subject to refinement as component selections and mechanical details mature.

11.2 Center of Mass Estimate

A one-dimensional center-of-mass model was created using the longitudinal position and mass of each internal spacecraft component.

The center of each component was calculated using:

$$\text{Component Center} = \text{Start Position} + \text{Length} / 2$$

The spacecraft center of mass was then estimated using:

$$x_{\text{COM}} = \Sigma(m_i x_i) / \Sigma(m_i)$$

Using the current internal stack, the calculated internal-subsystem center of mass was approximately:

125.93 mm from the Gecko end

The geometric center of the 227 mm spacecraft length is:

113.5 mm

Therefore, the internal subsystem distribution produces an offset of approximately:

+12.43 mm

toward the avionics end of the spacecraft.

When the approximately **390 g structure** is included and assumed to be centered at the geometric midpoint, the combined preliminary spacecraft center of mass becomes approximately:

123.55 mm

This corresponds to an offset of approximately:

+10.05 mm from the geometric center

11.3 Mass-Properties Interpretation

The preliminary COM result is considered acceptable for continued design development, but it should not yet be treated as the final spacecraft center of mass.

The current model does not fully include the mass and exact locations of:

- deployable solar panels

- hinge assemblies
- external antennas
- detailed structural members
- fasteners
- wiring distribution
- thermal hardware

These components could shift the final center of mass.

The current result is therefore classified as:

PASS prelim

for continued architecture and ADCS analysis.

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12. Final Preliminary Configuration

The current AURORA-1 baseline is a **2U Earth-observation CubeSat** designed around a 500 km circular low-Earth orbit and a compact visible-light imaging payload. The configuration integrates the payload, EPS, onboard computer, ADCS, UHF communications, S-band payload

communications, deployable solar arrays, antennas, and supporting structural and thermal concepts within the selected 2U architecture.

The design was refined through requirements development, subsystem trade studies, MATLAB simulations, Excel-based engineering budgets, and Autodesk Fusion CAD packaging. The current CAD model, **AURORA-1_2U_Layout_v1.0**, represents the first frozen preliminary mechanical baseline for the project.

The current configuration should be considered a **preliminary digital design baseline** rather than a manufacturing-ready spacecraft. Major system-level design decisions have been made, but several subsystem details remain open for higher-fidelity analysis and final verification.

13. Risks and Future Work

Although the current preliminary analyses indicate that the AURORA-1 architecture is feasible, several technical risks and open items remain.

The highest-priority future work includes:

- completing orbital lifetime and atmospheric-drag analysis for the six-month mission requirement
- refining ground-station access and communications-pass analysis
- verifying final transmitter, receiver, and antenna performance using vendor-specific specifications
- completing detailed EPS rail and channel allocation
- refining solar-cell packing, orientation, degradation, and temperature losses
- developing a higher-fidelity multi-node thermal model
- performing local payload heater sizing instead of spacecraft-level lumped heating
- expanding the ADCS simulation to full three-axis dynamics
- evaluating reaction-wheel saturation and momentum unloading
- including deployed solar-array and antenna masses in the final center-of-mass model
- refining structural geometry, brackets, fasteners, harness routing, and deployment hardware
- completing mission-level requirements traceability and system architecture documentation

Future development could also include higher-fidelity orbital simulation, finite-element structural analysis, detailed thermal analysis, and eventual hardware prototyping. These activities are intentionally outside the current digital preliminary-design scope.

14. Conclusion

The AURORA-1 project demonstrates the preliminary systems engineering and spacecraft design of a 2U Earth-observation CubeSat. The design evolved from an initial 1U concept into a 2U architecture after mass, volume, payload, and subsystem trade studies showed that additional spacecraft volume and design margin were necessary.

First-pass engineering analyses established a nominal 500 km orbit, approximately 94.6-minute orbital period, 3.59 W average spacecraft power demand, 7.5 W solar-array generation target, 100 kbps minimum payload downlink requirement, and $\pm 2^\circ$ imaging-pointing requirement. Preliminary communications, thermal, ADCS, power, mass-property, and CAD analyses indicate that the selected architecture is feasible within the assumptions of the current models.

AURORA-1 remains a preliminary digital spacecraft design rather than a flight-qualified system. However, the project establishes a consistent engineering baseline connecting mission objectives, requirements, subsystem selections, analytical models, and mechanical architecture. The resulting design provides a foundation for higher-fidelity analysis, detailed subsystem development, and future verification activities.